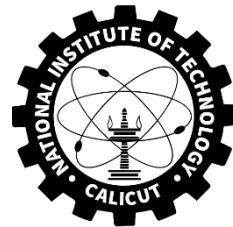


Quality, Reliability & Maintenance

ME3327E



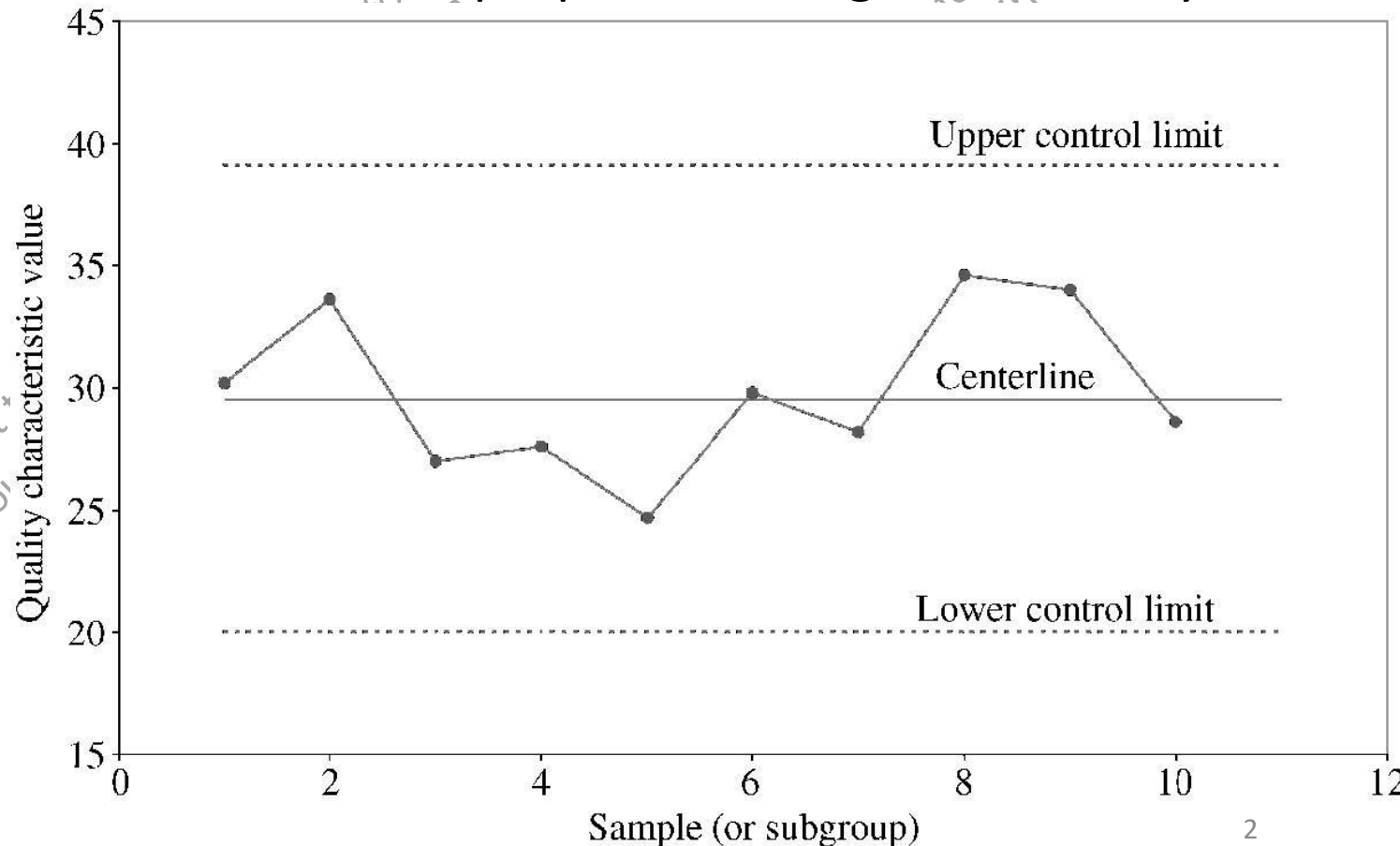
Course Instructor:

Dr. Nehem Tudu

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Statistical Process Control

- A control chart is a graphical tool for monitoring the activity of an ongoing process.
- Control charts are sometimes referred to as Shewhart control charts, because Walter A. Shewhart first proposed their general theory.



Benefits of Control Charts

- When to take corrective action.
- Type of remedial action necessary.
- When to leave a process alone.
- Process capability.
- Possible means of quality improvement.

Cause of Variations:

- Special Cause (*As per Deming 15%*): If an observation falls outside the control limits or a non-random pattern is exhibited, special causes are assumed to exist, and the process is said to be out of control.
- Common Cause (*As per Deming 85%*): If quality characteristic values are within control limits and no non-random pattern is visible, it is assumed that a system of common causes exists and that the process is in a state of statistical control.

Selection of Control Limits

The center line (CL) and control limits (UCL/LCL) are given by:

$$CL = E(\hat{\theta})$$

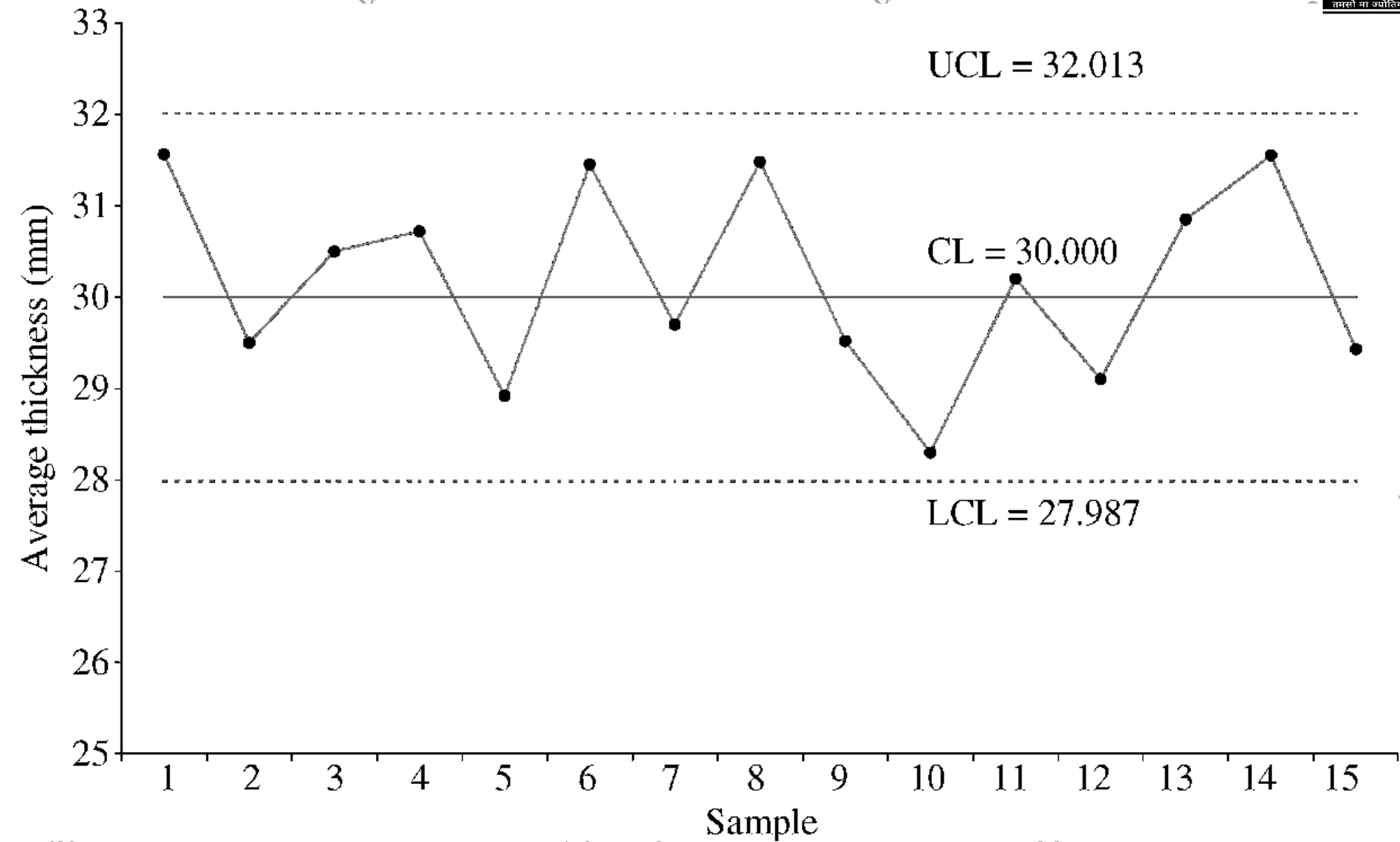
$$UCL = E(\hat{\theta}) + k\sigma(\hat{\theta})$$

$$LCL = E(\hat{\theta}) - k\sigma(\hat{\theta})$$

Exercise: A semi automatic turret lathe machines the thickness of a part that is subsequently used in an assembly. The process mean is known to be 30 mm with a standard deviation of 1.5mm. Construct a control chart for the average thickness using 3σ limits if samples of size 5 are randomly selected from the process. Table shows the average thickness of 15 samples selected from the process. Plot these on a control chart and make inferences.

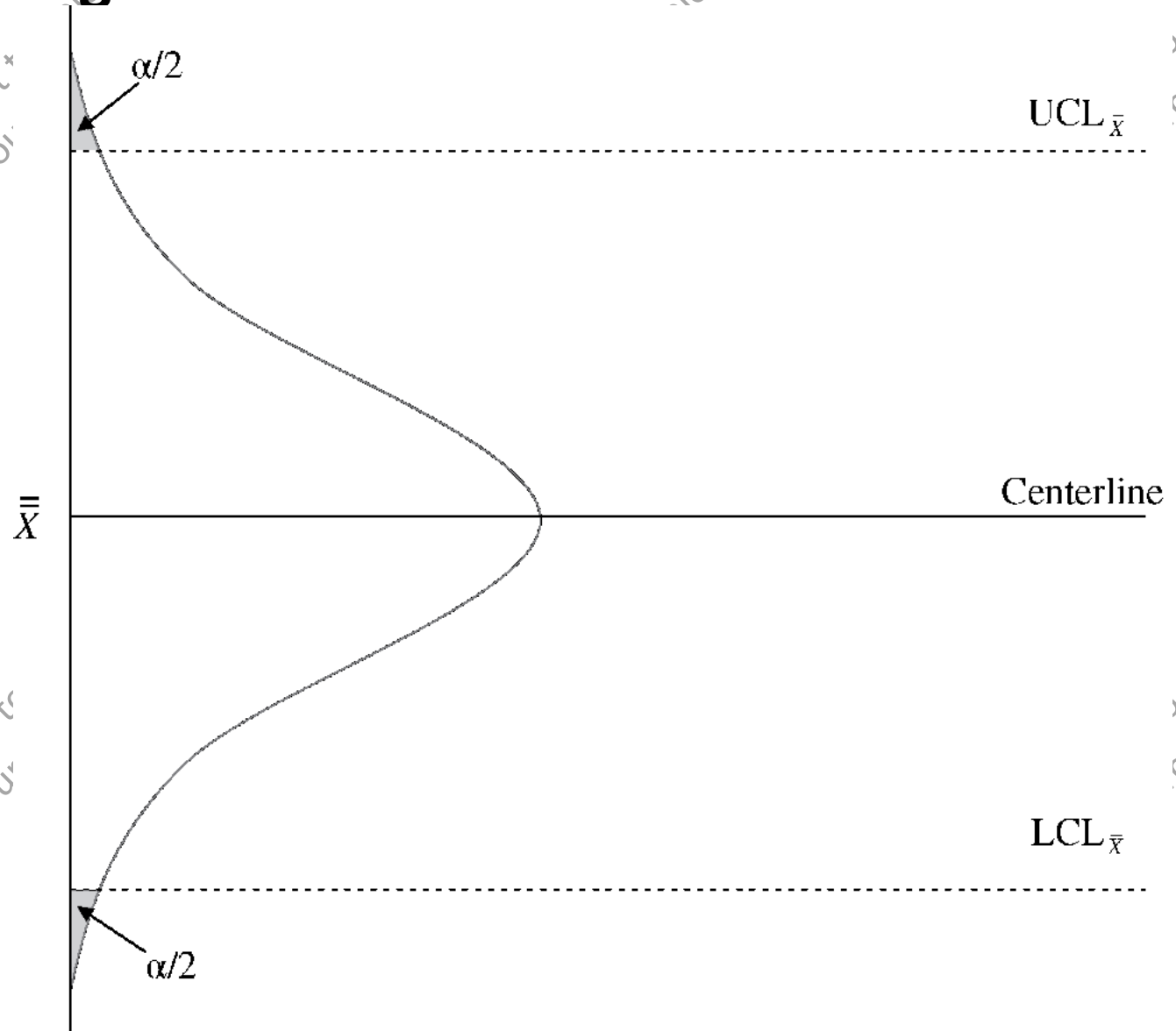
Sample	$\bar{x}$ (mm)	Sample	$\bar{x}$ (mm)	Sample	$\bar{x}$ (mm)
1	31.56	6	31.45	11	30.20
2	29.50	7	29.70	12	29.10
3	30.50	8	31.48	13	30.85
4	30.72	9	29.52	14	31.55
5	28.92	10	28.30	15	29.43

Control chart for average thickness.



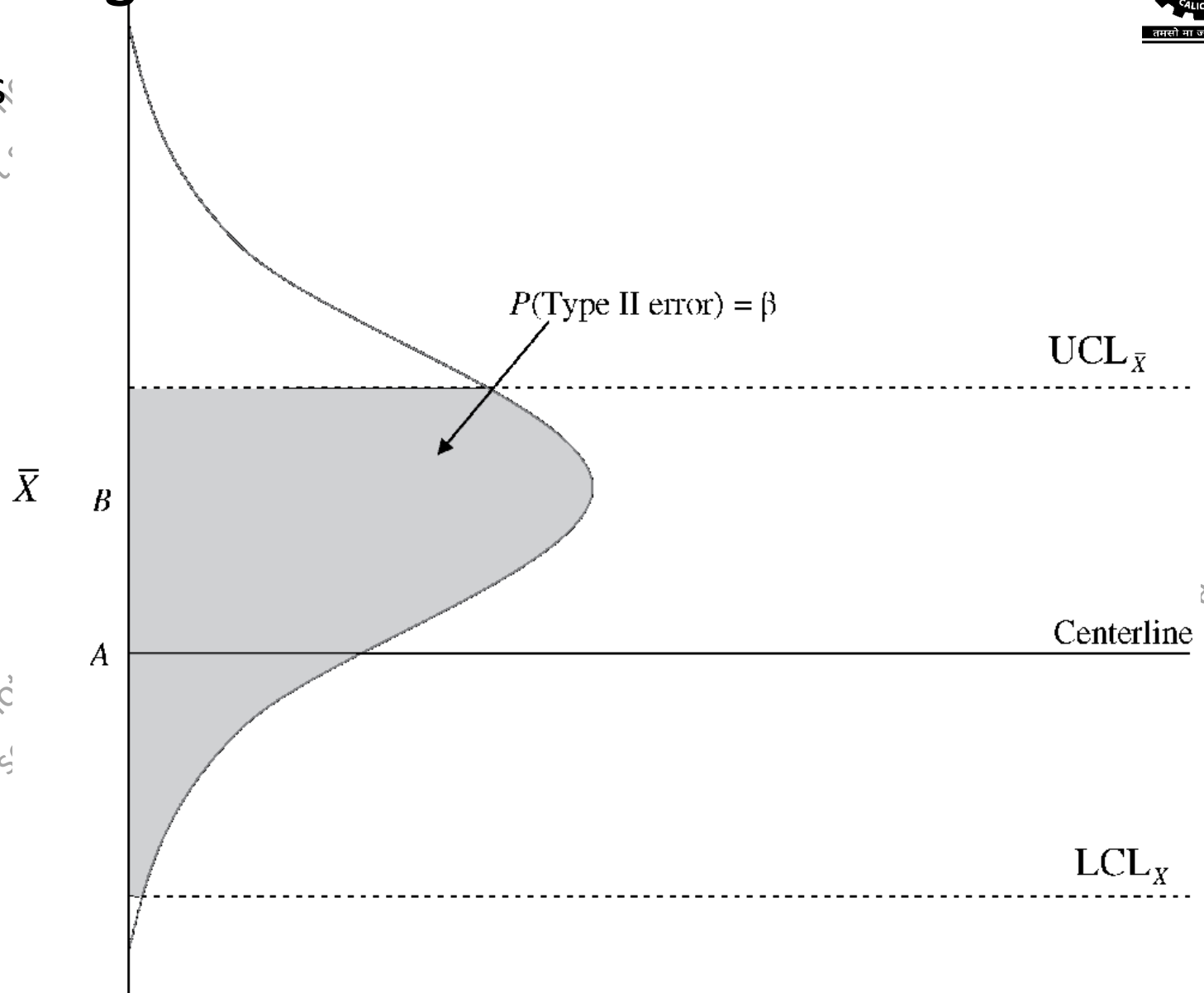
Errors in Making Inferences from Control Charts

- Type I Errors



Errors in Making Inferences from Control Charts

- Type II Errors

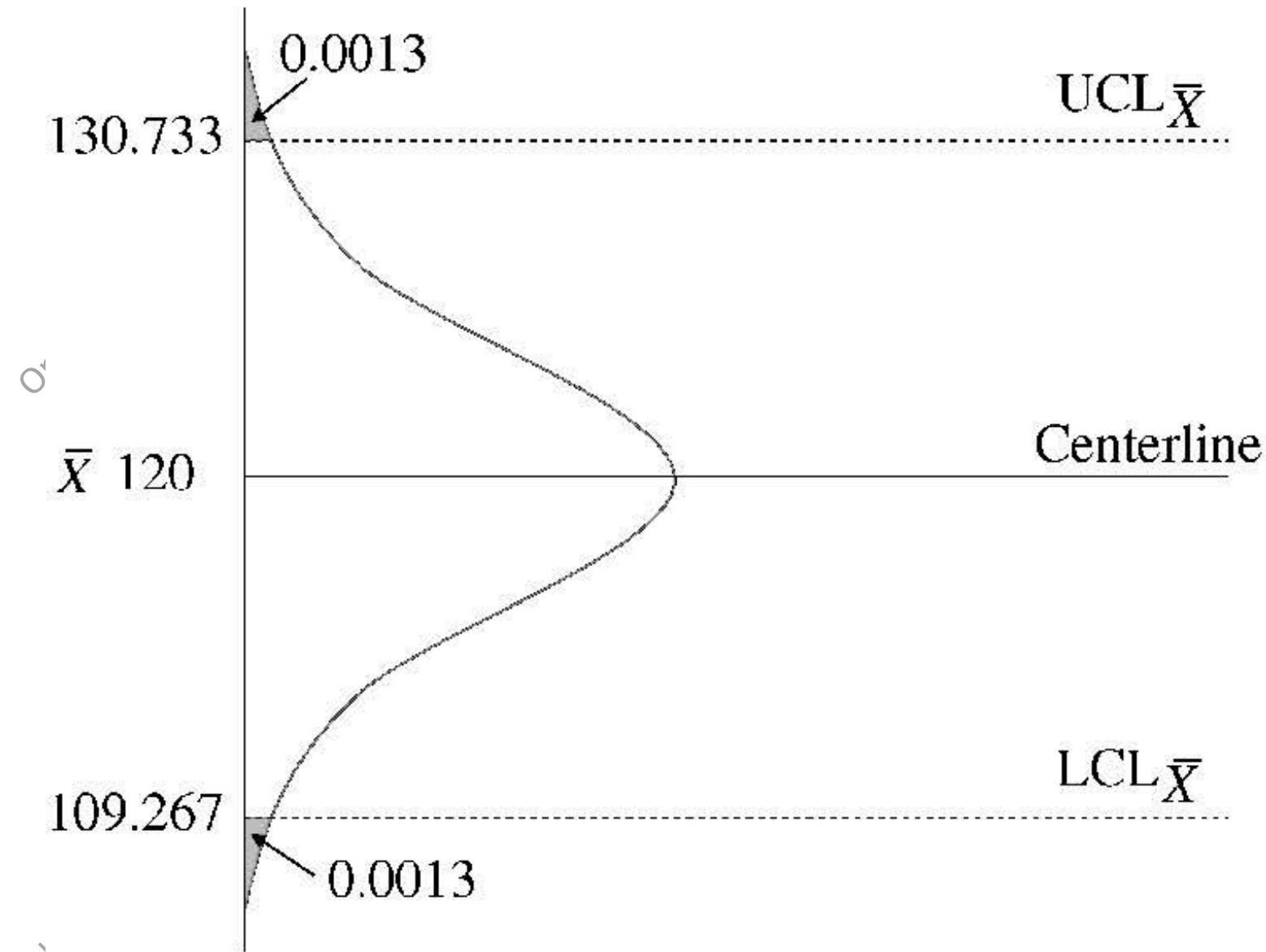


Exercise 2

A control chart is to be constructed for the average breaking strength of nylon fibers. Samples of size 5 are randomly chosen from the process. The process mean and standard deviation are estimated to be 120 kg and 8 kg, respectively.

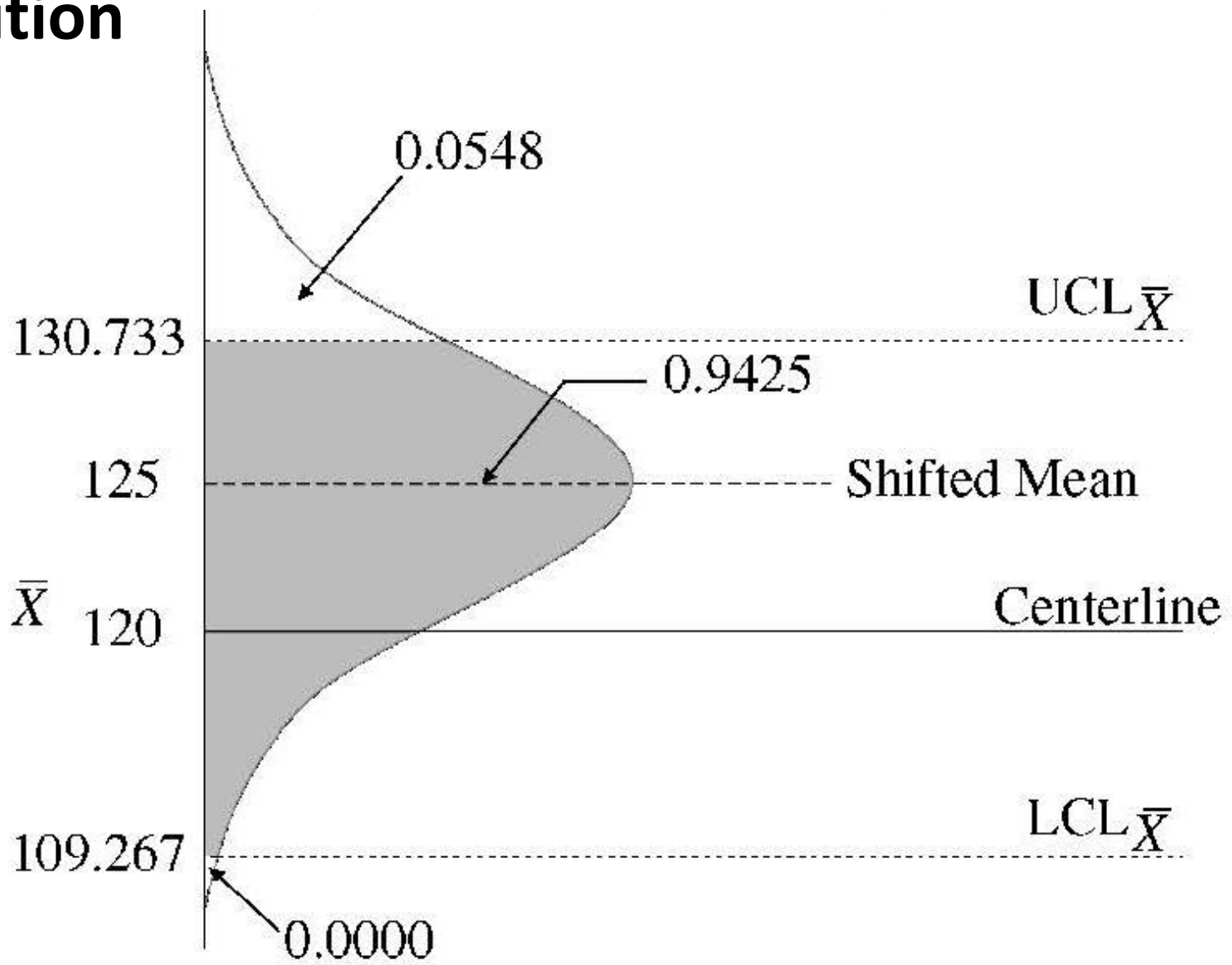
- If the control limits are placed three standard deviations from the process mean, what is the probability of a type I error?
- If the process mean shifts to 125 kg, what is the probability of concluding that the process is in control and hence making a type II error on the first sample plotted after the shift?
- What is the probability of detecting the shift by the second sample plotted after the shift if the samples are chosen independently?

Solution



(a) Probability of a type I error

Solution



(b) Probability of a type II error